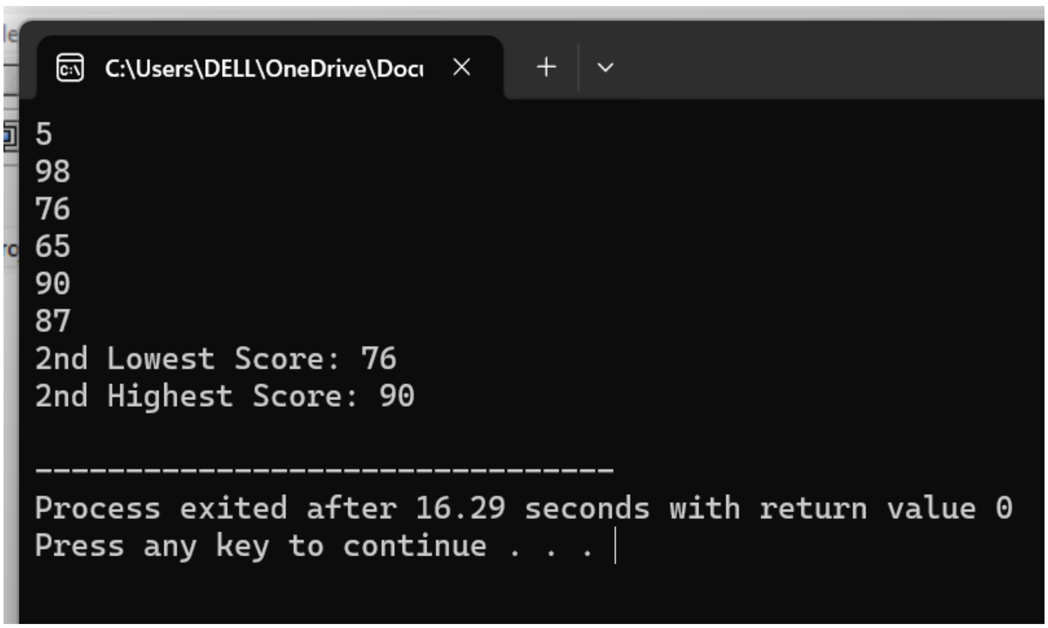
PROGRAM NO:01

```
#include <iostream>
using namespace std;
int calculate2Low(int arr[], int size)
{
    int lowest = arr[0];
    for (int i = 1; i < size; i++)
    {
        if (arr[i] < lowest)
        {
            lowest = arr[i];
        }
    }

    int secondLowest = 1000000;
    for (int i = 0; i < size; i++)
    {
        if (arr[i] > lowest && arr[i] < secondLowest)
        {
            secondLowest = arr[i];
        }
```

```
    }

    return secondLowest;
}
int calculate2High(int arr[], int size)
{
    int highest = arr[0];
    for (int i = 1; i < size; i++)
    {
        if (arr[i] > highest)
        {
            highest = arr[i];
        }
    }

    int secondHighest = -1000000;
    for (int i = 0; i < size; i++)
    {
        if (arr[i] < highest && arr[i] > secondHighest)
        {
            secondHighest = arr[i];
```

```
        }
    }

    return secondHighest;
}

int main()
{
    int N;

    cin >> N;

    int scores[N];

    for (int i = 0; i < N; i++)
    {
        cin >> scores[i];
    }

    cout << "2nd Lowest Score: " << calculate2Low(scores,
N) << endl;
```

```
        cout << "2nd Highest Score: " << calculate2High(scores, N) << endl;

        return 0;
    }
```

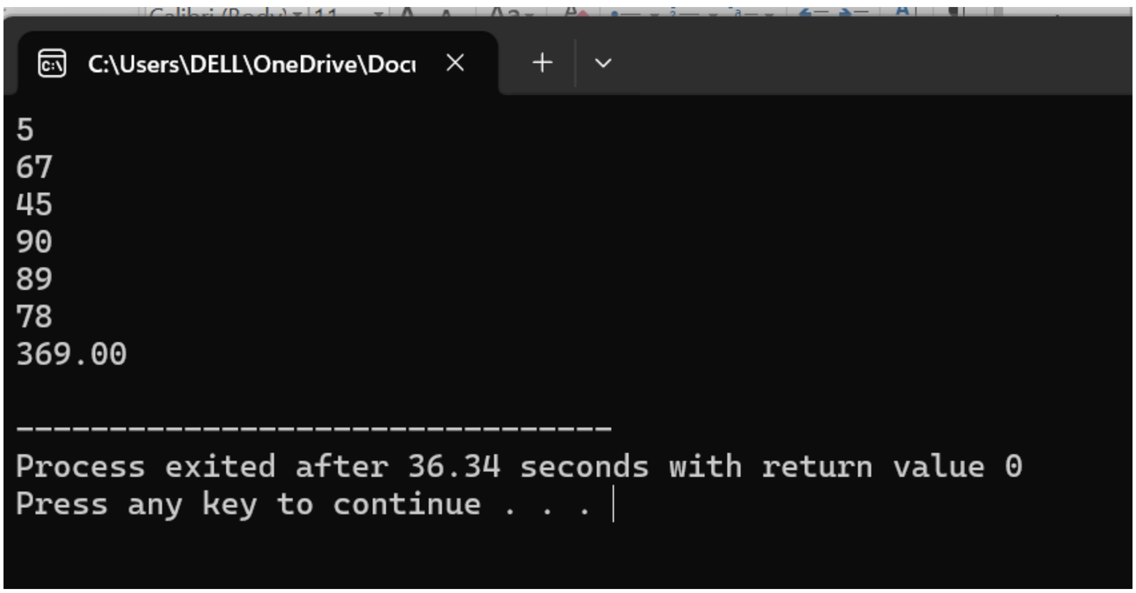


PROGRAM NO:02

```
#include <iostream>
#include <iomanip>
using namespace std;



double calculateSum(int arr[], int size)
```

```
{
    int sum = 0;

    for (int i = 0; i < size; i++)
    {
        sum += arr[i];
    }

    return sum;
}

int main()
{
    int N;
    cin >> N;

    int scores[N];

    for (int i = 0; i < N; i++)
    {
        cin >> scores[i];
```

```
    }

    double total = calculateSum(scores, N);

    cout << fixed << setprecision(2) << total << endl;

    return 0;
}
```



PROGRAM NO:03

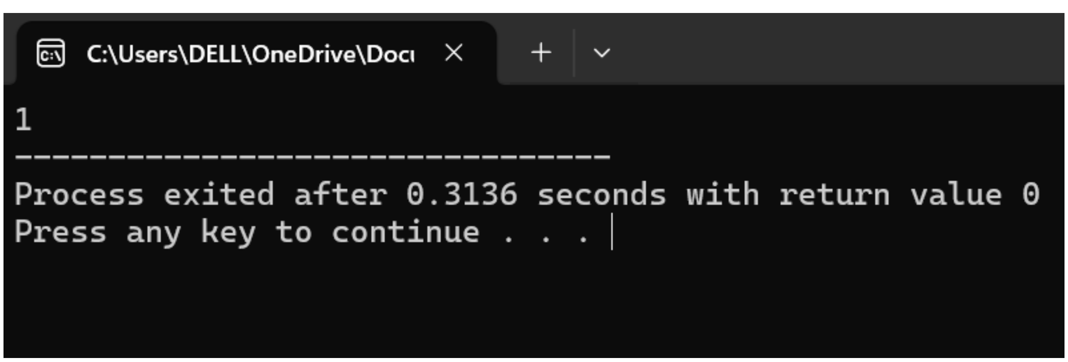
```
#include <iostream>
#include <iomanip>
using namespace std;
double celsiusToFahrenheit(double celsius)
```

```
{
    return (celsius * 9.0 / 5.0) + 32;
}

int main()
{
    double celsius;

    cin >> celsius;

    double fahrenheit = celsiusToFahrenheit(celsius);

    cout << fixed << setprecision(1) << fahrenheit << endl;

    return 0;}
```

```
C:\Users\DELL\OneDrive\Docı
45
113.0

--------------------------------
Process exited after 20.28 seconds with return value 0
Press any key to continue . . .
```

PROGRAM NO:04

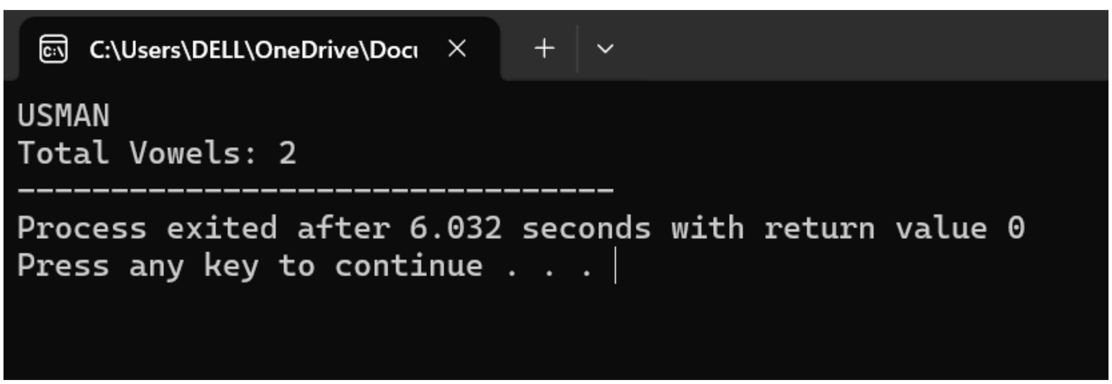
```
#include <iostream>
using namespace std;

bool isLeapYear(int year) {
   if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
{
       return true;
   }
   return false;
}

int main() {
   cout << isLeapYear(2024);
   return 0;
}
```



PROGRAM NO:05

```
#include <iostream>
using namespace std;

int countVowels(char word[])
{
    int count = 0;

    for (int i = 0; word[i] != '\0'; i++)
    {
        if (word[i] == 'a' || word[i] == 'e' || word[i] == 'i' ||
            word[i] == 'o' || word[i] == 'u' ||
            word[i] == 'A' || word[i] == 'E' || word[i] == 'I' ||
            word[i] == 'O' || word[i] == 'U')
        {
            count++;
        }
    }

    return count;
}
```

```
int main()
{
    char name[100];

    cin >> name;

    cout << "Total Vowels: " << countVowels(name);

    return 0;
}
```



PROGRAM NO:06

```
#include <iostream>
using namespace std;
```

```
int main()
{
    char arr[5];
    char temp;

    for (int i = 0; i < 5; i++)
    {
        cin >> arr[i];
    }
    for (int i = 0; i < 4; i++)
    {
        for (int j = 0; j < 4 - i; j++)
        {
            if (arr[j] > arr[j + 1])
            {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
```

```
    cout << "Sorted Letters: ";
    for (int i = 0; i < 5; i++)
    {
        cout << arr[i] << " ";
    }

    return 0;
}
```